Parachute Gliding System (C++)

A highly detailed **parachute and gliding system for Unreal Engine 5**, inspired by the parachuting mechanics found in modern battle royale games such as **PUBG**. The system recreates the complete aerial gameplay loop, from the initial parachute deployment to controlled gliding, landing, and player movement in the air.

Built primarily in **C++**, the system focuses on responsive controls, realistic aerial movement, and the large number of gameplay details required to make parachuting feel polished and consistent.

Key Features

- **Parachute Deployment**
- Complete parachute deployment and opening sequence.
- Transition from freefall into parachute gliding.
- Configurable deployment conditions and gameplay parameters.

- Smooth transitions between different aerial states.

- **Freefall System**

- High-speed vertical freefall before parachute deployment.
- Player-controlled aerial movement during the initial descent.
- Dynamic control over horizontal direction and positioning.

- Seamless transition from freefall to parachuting.

- **Parachute Gliding**

- Full directional control while gliding.
- Forward, backward, and lateral movement.
- Dynamic aerial steering based on player input.

- Adjustable movement and glide parameters for different gameplay requirements.

- **Battle Royale-Style Controls**

- Designed around the familiar aerial movement found in modern battle royale games.
- Allows players to control both **direction and landing position** during descent.
- Supports tactical positioning before reaching the ground.

- Designed to integrate naturally with larger multiplayer battle royale systems.

- **Landing System**

- Controlled descent toward the selected landing area.
- Automatic transition from aerial movement to ground movement.
- Landing detection and state management.

- Designed to provide a smooth transition into normal player locomotion.

- **Camera & Player Feedback**

- Camera behavior changes according to the current aerial state.
- Visual feedback during freefall, parachute deployment, and gliding.
- Responsive camera movement designed to improve spatial awareness while airborne.

Technical Focus

The primary focus of this project is reproducing a **complete battle royale-style aerial movement system**, rather than implementing a simple parachute animation. Freefall, parachute deployment, aerial steering, gliding, landing detection, player states, and camera behavior are integrated into a unified gameplay system.

The system is built with **C++ architecture and configurable gameplay parameters**, making it suitable for integration into larger multiplayer projects.

It can serve as a foundation for **battle royale games, military shooters, open-world games, and other projects requiring controlled aerial traversal and parachute-based deployment mechanics**.